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Editorial

Plant Disease: A Growing Threat to Global Food Security

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Abstract: The escalating global population has led to an increased demand for both quantity and quality in food production. Throughout history, plant diseases have posed significant threats to agricultural output by causing substantial food losses annually while also compromising product quality. Accurate identification of pathogens, clarifying the pathogenic mechanism of pathogens, and understanding the interaction between pathogens and hosts are important for the control of plant diseases. This Special Issue, "Research Progress on Pathogenicity of Fungi in Crops", belongs to the section "Pest and Disease Management" of *Agronomy*. It contains research papers on the identification and phylogeny of fungal pathogens, the molecular genetics of plant fungal pathogens, the molecular mechanisms of fungal pathogenicity, and the molecular basis of the interaction between fungi and crops. These studies encapsulate efforts to understand disease systems within current genomics, transcriptomics, proteomics, and metabolomics studies, highlighting research findings that could be future targets for crop disease and pest control. The studies presented in this Special Issue promote the progress of fungal pathogenicity research in crops and provide a scientific basis for future disease control, which is of great significance for sustainable agricultural development and global food security.

Keywords: fungal pathogenicity; plant–pathogen interactions; plant growth-promoting rhizobacteria (PGPR); biological control; comparative genomics; disease management; crop protection

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1. Introduction

Plant diseases refer to a series of abnormal physiological processes caused by the infection of plants by various pathogens, such as fungi, bacteria, and viruses, resulting in the retardation of plant growth and development, a decrease in yield and quality, and even death in severe cases [1,2]. According to the type of pathogen, plant diseases can be divided into fungal diseases, bacterial diseases, viral diseases, and nematode diseases [3]. The occurrence and prevalence of plant diseases are affected by many factors, including the species and varieties of host plants, the pathogenicity and virulence of pathogens, environmental conditions such as temperature and humidity, light, and soil type, and human factors such as cultivation management measures [4]. Pathogens are mainly transmitted between plants through wind, rain, insects, seeds, farm tools, and other methods. They can also survive over winter through soil residues, intermediate hosts, and other methods, thus causing the periodic occurrence of diseases [5]. Factors such as climate change, changes in farming practices, loss of resistant varieties, and reduced biodiversity exacerbate the risk of plant disease outbreaks [6,7].

Plant diseases pose a serious threat to agricultural production and food security. The annual loss of crop yield due to plant diseases is estimated to be more than 30% globally, worth hundreds of billions of dollars [8,9]. Diseases not only directly lead to crop reduction but also reduce the quality and value of agricultural products, increase agricultural inputs and labor costs, and affect farmers' income and livelihood [10]. In addition, some plant pathogens (such as *Fusarium graminearum* and *Aspergillus flavus*) can also produce toxic

secondary metabolites, polluting agricultural products and endangering human and animal health [11]. With the continuous growth of the world's population and the decline in agricultural productivity, food supply is under great pressure, and the problem of plant diseases has become increasingly prominent, which poses a major challenge to global food security [6,12].

Chemical control has long been the mainstay of plant disease management. However, excessive dependence on pesticides not only results in ecological problems such as environmental pollution and loss of biodiversity but also leads to the generation and aggravation of pathogen resistance, reducing the control effect [13,14]. Therefore, there is an urgent need to develop more environmentally friendly, safe, and sustainable integrated control strategies and technologies for plant diseases. The breeding of disease-resistant varieties is the basis and key for controlling plant diseases. Through conventional breeding or molecular breeding technology, it is possible to cultivate and promote excellent varieties with disease resistance and disease tolerance, reduce pesticide use, and improve crop yield and quality [15,16]. Moreover, the agricultural sector should strengthen plant quarantine and disease monitoring and early warning, detect and eliminate the hidden dangers of epidemics in a timely manner, and curb the spread of disease [17].

Biocontrol is a method that uses organisms or products from biological sources to suppress or eliminate pests and has the advantages of being environmentally friendly, pollution-free, and sustainable [18,19]. With the increase in people's awareness of environmental protection and the promotion of sustainable agriculture, the importance of biological control in the integrated management of crop pests and diseases has become increasingly prominent [20]. Microorganisms are widely distributed in nature and include a wide variety of species, some of which have strong antagonistic effects and can inhibit the growth and reproduction of pathogenic bacteria; therefore, they can be used as potential biological control agents in agricultural production. At present, many microorganisms (such as antagonistic fungi and bacteria) have been developed into commercial biocontrol agents and applied to the control of plant diseases, and remarkable results have been achieved [21]. However, the large-scale application of biocontrol agents still faces many challenges, such as unstable control effects, high production costs, short shelf life, and a lack of regulatory and policy support [22,23]. It is necessary to further explore and utilize biocontrol microbial resources, improve the key technologies of industrialization, establish and improve management systems, and promote biological control to play a greater role in the sustainable prevention and control of plant diseases.

This Special Issue of *Agronomy*, "Research Progress on Pathogenicity of Fungi in Crops", collected original contributions mainly addressing pathogenic fungal identification and phylogeny, the molecular genetics of plant fungal pathogens, fungal pathogenic molecular mechanisms, the molecular basis of fungal–crop interactions, omics studies (genomics, transcriptomics, proteomics, and metabolomics), and disease and pest management. These studies provide new insights into our deeper understanding of the pathogenic mechanisms of fungal pathogens and their interactions with crops and lay the foundation for the development of new disease control strategies.

2. Overview of Published Articles

This special issue contains several original research articles concerning the pathogenicity and control of fungal diseases in various crops. Fungal diseases can affect crops such as coconut palm, rice, soybean, maize, tomato, grapevine, apple, and banana.

2.1. Biological Control and Antimicrobial Activity

Biocontrol plays a crucial role in sustainable agriculture, especially in managing plant diseases and pests, providing a feasible alternative to reduce reliance on chemical pesticides [24,25]. In recent years, there has been an increasing focus on the potential of beneficial microorganisms and their metabolites for crop protection [26,27]. These innovative studies have established a solid foundation for the development of efficient and

eco-friendly biocontrol strategies, bringing new hope for the achievement of sustainable agricultural development [28]. This Special Issue highlights several recent advancements in biocontrol and antimicrobial activity, focusing on the potential of beneficial microorganisms such as *Bacillus velezensis*, *Bacillus subtilis* and *Trichoderma echinosporum* to suppress plant pathogens and enhance crop growth [29–31]. Through the comprehensive use of genomics, comparative genomics, transcriptomics, and other multiomics technologies, researchers have investigated the genetic characteristics and secondary metabolite production of biocontrol strains. These studies provide essential theoretical guidance for the development and application of optimized biocontrol agents.

Li et al. studied the antibacterial activity of *Bacillus velezensis* Htq6 isolated from walnuts and its inhibitory effect on gray mold germ [30]. This study explored the biological control mechanism of this strain and its potential in plant disease control through genome sequencing, comparative genomics, and transcriptomic analysis. Through antiSMASH (v5.0.0) software analysis, 14 secondary metabolite gene clusters were predicted in the Htq6 strain, including Plantazolicin, Macrolactin, and Bacillaene. *B. velezensis* Htq6 has significant antibacterial activity and inhibits the growth and reproduction of *Botrytis cinerea* by interfering with its energy metabolism, cell wall synthesis, and antioxidant mechanisms. The lipopeptide compounds produced by Htq6 act on the cell wall and plasma membrane of *B. cinerea*, interfering with its structure, leading to the accumulation of intracellular reactive oxygen species (ROS), and destroying the normal physiological metabolism and growth rate of Botrytis cinerea. These findings provide a theoretical basis for the further development of Htq6 as a biological control agent, showing the potential application of this strain in the biological control of plant diseases, and Htq6 is expected to become an effective biological control agent.

Hao et al. studied the ability of the 6-pentyl-α-application of pyranone (6-PP) to inhibit the growth and development of *Fusarium oxysporum* hf-26 in a tomato soilless culture system [29]. 6-PP is a volatile organic compound (VOC) secreted by *Trichoderma echinosporum* that is known to have a wide range of antibacterial activities. The main purpose of this study was to evaluate the ability of 6-PP to inhibit Verticillium wilt in tomato plants and explore its potential application in soilless culture nutrient solutions. These results demonstrate that, at relatively high concentrations, 6-PP significantly inhibits the mycelial growth of *F. oxysporum* hf-26, suggesting its potential as a biological control agent in tomato soilless culture systems. Further study validated the antibacterial mechanism and practical application of 6-PP through gene expression analysis and greenhouse experiments, providing a scientific basis and technical support for its future application in agricultural production. These findings indicate that 6-PP, a volatile organic compound secreted by *T. echinosporum*, effectively reduces the occurrence of tomato Verticillium wilt by inhibiting the mycelial growth and toxin synthesis of *F. oxysporum*. This study provides a scientific basis for the application of 6-PP in soilless cultivation systems with potential benefits, including reduced use of chemical pesticides and decreased environmental pollution. By exploiting the natural antimicrobial properties of beneficial microorganisms and their metabolites, we can develop effective and environmentally friendly strategies for plant disease management.

Shang et al. focused on the effect of *Bacillus subtilis*, a Gram-positive, rod-shaped bacterium, on promoting the growth of cigar tobacco and improving the soil microbial structure, particularly in the control of the root black rot pathogen *Thielaviopsis basicola* [31]. Plant growth-promoting rhizosphere bacteria (PGPRs) are important beneficial species known for their ability to promote plant growth, inhibit harmful pathogens, and synthesize compounds that directly affect plant morphogenesis or activate ineffective soil components [32]. In this study, a controlled pot experiment was performed to investigate the mechanism by which rhizosphere probiotics repair the soil ecosystem and promote the growth of cigar tobacco plants infected with tobacco root black rot. The results revealed that the presence of pathogenic bacteria worsened the soil environment, limited the conversion of soil nutrients and the absorption and utilization of nutrients by plants, and

seriously hindered the growth and development of cigar tobacco plants. In contrast, the application of PGPR microbial fertilizer changed the soil microbial community structure and antagonized indigenous pathogen populations. Compared with the CK treatment, the application of PGPR microbial fertilizer significantly improved the catalytic ability of soil enzymes and the activities of peroxidase, acid phosphatase, urease and sucrase. These improvements in soil parameters, including plant height, stem circumference, number of leaves, maximum leaf size, and fresh and dry weight, further contributed to the growth of the cigar tobacco plants.

Degani and Gordani investigated the efficacy of 6-pentyl-α-pyrone (6-PP), an important antifungal compound secreted by *Trichoderma asperellum*, against the maize late wilt disease (LWD) pathogen *Magnaporthiopsis maydis* [33]. This disease poses a serious threat to maize production in Israel and Egypt. Although 6-PP shows extremely high potential for use in plate experiments, it has not yet been tested for plant treatment prevention. In controlled growth chamber experiments, direct application of *T. asperellum* to maize seeds at sowing provided significant protection to young sprouts for up to 42 days. This treatment resulted in more than twofold growth promotion and reduced root infection by *M. maydis*, as detected by real-time PCR. However, when the same procedure was tested in a commercial field setting, the protective effects were less pronounced. Despite this, the field trials showed an 11% reduction in cob symptoms and a ninefold decrease in the amount of pathogen DNA in the stem tissue, indicating a considerable level of disease control. This study suggested that the biocontrol efficacy of *T. asperellum* is largely due to its secreted metabolites, particularly 6-PP. The application of 6-PP in seed coatings significantly reduced pathogen infection and promoted plant growth recovery. This research revealed that 6-PP has considerable potential as a biofriendly antifungal treatment for maize late wilt disease. These findings offer valuable theoretical support and practical application prospects for the future development of biopesticides and plant vaccines based on 6-PP, which are anticipated to have widespread use in agricultural production, reduce reliance on chemical pesticides, and promote the sustainable development of agriculture while protecting ecological environments.

2.2. Pathogen Interactions and Defense Mechanisms

The interaction between plants and pathogens is a complex and dynamic process involving the interaction of multiple defense mechanisms and pathogenic agents [34,35]. Understanding plant–pathogen interactions is critical for the development of novel, efficient, and sustainable plant disease control strategies. In recent years, with the rapid development of omics technologies, an increasing number of studies have focused on elucidating the molecular mechanisms of plant–pathogen interactions, which has opened up new horizons for the study of plant immunology. In this review, we summarize several recent research advances on plant–pathogen interactions and defense mechanisms, focusing on the pathogenic mechanisms of important plant pathogens and the resistance mechanisms of their host plants.

Liang et al. focused on the mechanism of *Alternaria alternata* to antioxidant molecules and its role in plant pathogenicity [36]. This study used wild-type *A. alternata* Z7 and mutant strains and revealed the regulatory mechanism of antioxidant system-related genes in response to reactive oxygen species (ROS) stress through comparative transcriptome analysis. These genes are closely related to antioxidant defense and ROS detoxification. This study revealed that antioxidant molecules (including enzyme scavengers such as superoxide dismutase, catalase, thioredoxin and glutathione) and nonenzymatic compounds play crucial roles in maintaining the redox homeostasis of *A. alternata* and regulating its pathogenesis. Comparative transcriptomics of antioxidant mutants revealed significant effects on glutathione metabolism, cell detoxification, osmotic stress response, and secondary metabolic gene expression, highlighting the complex and interrelated regulatory network involved in fungal antioxidant defense.

Huang et al. provided a comprehensive examination of the role of ACT toxin in the pathogenicity of the *Alternaria alternata* tangerine pathotype and its impact on various citrus species [37]. Host-specific toxins (HSTs) play important roles in the process of pathogen infection. The biosynthesis of toxins involves multiple genes, most of which are located on conditionally separable chromosomes (CDCs). The results showed that different citrus varieties had different sensitivities to toxins, which provides an important theoretical basis for further study of the biosynthetic mechanism of the ACT toxin and the development of control strategies against *A. alternata* infection. The findings indicate that citrus varieties sensitive to ACT toxin displayed noticeable necrotic lesions, while those resistant to ACT toxin, showed no obvious lesions. These results suggest that the tolerance of citrus plants to ACT toxin is a key factor in their ability to resist brown spot disease. These findings provide a new perspective for further understanding the pathogenic mechanism of *A. alternata* and provide an important theoretical basis for the development of disease-resistant citrus varieties.

Divino Rosa dos Santos Junior et al. investigated a critical issue in agricultural production: the susceptibility of popcorn genotypes to foliar diseases, which can lead to substantial economic losses [38]. The authors conducted a comprehensive evaluation of 15 S7 inbred lines of popcorn and their respective testcross hybrids. The main diseases studied included leaf surface diseases caused by *Puccinia polysora*, *Exserohilum turcicum*, and *Bipolaris maydis*. Results demonstrated that different genotypes exhibit significant genetic variation in terms of agronomic traits and disease resistance, providing a crucial reference for breeding high-yield and disease-resistant hybrids. The study's methodology and findings can serve as a model for similar research in other crops, emphasizing the importance of genetic diversity and hybrid vigor in plant breeding programs. This research offers valuable insights into genetic improvement in popcorn. By demonstrating the effectiveness of allelic complementation in enhancing hybrid resistance to foliar diseases, this study paves the way for the development of robust, high-yielding popcorn cultivars.

2.3. Genetic Diversity and Population Structure

Genetic diversity and population structure are important components of plant pathology research, and an in-depth understanding of the genetic variation and adaptive mechanism of pathogen populations is crucial for clarifying disease prevalence and optimizing prevention and control strategies [39]. In recent years, with the rapid development of molecular marker technology, an increasing number of studies have focused on the use of genomics, population genetics, and other means to reveal the genetic diversity and evolutionary dynamics of plant pathogens [40]. These studies not only deepen our understanding of the adaptive evolution of pathogens but also provide new ideas for the development of accurate and efficient plant disease prevention and control measures.

Liu et al. focused on the genetic variation and adaptability of *Colletotrichum gloeosporioides* in different regions and different citrus varieties in China by analyzing the genetic structure and diversity of the population of citrus plants [41]. This study selected 63 *C. gloeosporioides* strains from 8 different regions and 5 different citrus varieties in China and conducted genetic diversity analysis using glyceraldehyde-3-phosphate dehydrogenase (GAPDH) gene sequences. A total of 19 GAPDH haplotypes were identified, of which the main haplotype (H5) was widely distributed among 28 strains, mainly from Huangyan (HY), Linhai (LH) and Jiande (JD) in Zhejiang Province and Citrus varieties from Mengshan (GX), Guangxi Province. The results showed that there was rich haplotype diversity in the *C. gloeosporioides* population, and the main haplotype, H5, was widely distributed in the study area, indicating that *C. gloeosporioides* may spread through asexual reproduction and gene flow in these areas. Phylogenetic analysis revealed that the genetic variation of *C. gloeosporioides* is not related to geographical origin or host and is mainly caused by gene mutation, recombination, or migration. This finding provides an important basis for further research on the genetic variation mechanism of *C. gloeosporioides*. These findings provide an important basis for further research on the genetic variation mechanism and adaptability of

C. gloeosporioides and provide a scientific reference for formulating prevention and control strategies for citrus diseases.

Vukotić et al. focused on the morphological, molecular, and pathogenic characterization of *Neofabraea alba*, a postharvest pathogen of apple in Serbia [42]. The authors isolated the strains from stored apples in 2017–2018 and studied 25 representative strains. It was confirmed by artificial inoculation experiments that these strains were pathogenic to both apple fruit and apple tree branches and could cause typical symptoms of white spot disease, and there were differences in virulence among the different strains. Phylogenetic analysis was performed using the internal transcriptional spacer (ITS) and 16S mitochondrial rRNA (mtSSU) gene sequences; these strains were identified as *Neofabraea alba*, and a certain genetic distance was found between the strains. The researchers divided the isolated strains into two main types from the perspective of morphology and thoroughly studied the influence of different media, temperature, pH and other environmental factors on the growth, morphology and conidial formation of these two types of strains. The results showed that environmental factors had a significant influence on the phenotype of the strains; for example, different media could affect colony color, mycelium density and conidial morphology. This study aimed to identify the causal agents of bull's eye rot in apples stored in Serbia during the period 2017–2018 and to characterize the obtained isolates based on pathogenicity, physiological morphological traits, and molecular criteria for academic research purposes.

2.4. Omics Technologies and Integrated Approaches

With the rapid development of modern biotechnology, multiomics technology has become an important means to analyze the mechanism of plant–pathogen interactions and guide disease prevention and control [43,44]. The article by Gai et al. presents a comprehensive study on the host interactions between *Alternaria alternata* and citrus plants [45]. The authors employed a comparative transcriptome approach to examine the dynamic defense transcriptome of *A. alternata* during citrus infection, aiming to reveal its pathogenic mechanism and strategy for coping with the host defense system. By inoculating citrus plants with the pathogen *A. alternata* and analyzing the resulting gene expression profiles, a substantial number of differentially expressed genes (DEGs) was identified. This study provides insights into the regulation of virulence genes in *A. alternata*, helping to identify key virulence factors that contribute to the ability of this pathogen to infect and cause disease in citrus plants. Through comparative transcriptomics analysis, this study revealed the dynamic changes in the gene expression of *A. alternata* during citrus infection, providing an important basis for understanding its pathogenic mechanism. These findings collectively enhance our understanding of the molecular mechanisms underlying *A. alternata* pathogenicity and provide valuable insights for developing targeted strategies to manage Alternaria brown spot in citrus crops.

Tzec-Sima et al. provided a comprehensive review of omics technologies used to manage diseases and pests affecting coconut palms (*Cocos nucifera* L.), an economically important crop in tropical regions [46]. This review highlights the challenges posed by diseases such as bud rot (caused by *Phytophthora palmivora*) and lethal yellowing (caused by various phytoplasmas), as well as pests such as the coconut palm weevil *Rhynchophorus vulneratus* (Coleoptera: Curculionidae) and the horned beetle *Oryctes rhinocerus* (Coleoptera: Scarabaeidae). The authors propose the integration of omics approaches, such as genomics, transcriptomics, proteomics, and metabolomics, to develop effective and sustainable pest and disease control strategies. These technologies can provide a comprehensive analysis of the biological systems of coconut palm and its pests, which is helpful for identifying genetic markers, biosynthetic pathways, and potential targets for intervention.

Chen et al. identified the PPO gene family of *Agaricus bisporus*, which is one of the most widely cultivated edible fungi in the world [47]. Polyphenol oxidase (PPO) is the critical enzyme in the enzymatic browning melanogenesis pathway of *Agaricus bisporus*. This paper identified six abppo family proteins, including AbPPO1, AbPPO2, AbPPO3,

AbPPO4, AbPPO5, and AbPPO6. A phylogenetic tree was constructed by employing the neighbor-joining (NJ) method in MEGA 7.0, which revealed that the PPOs could be divided into several distinct clades. This paper also studied the expression patterns of these genes at different growth stages, during postharvest storage and under abiotic stress. These systematic analyses provide the basis for further study of the biological function of the abppo gene in *B. bisporus*, and genetic and breeding analyses of related species provide a theoretical basis and reference.

2.5. Soil Health and Disease Prevention

Soil health and disease control are major challenges facing modern agriculture, and a thorough understanding of the relationship between soil factors and crop disease occurrence is crucial for the development of scientific, efficient and sustainable integrated prevention and control strategies [48,49]. In recent years, with the deepening of the understanding of soil ecosystems, an increasing number of studies have focused on exploring the effects of soil physicochemical properties, microbial composition, rhizosphere interactions and other factors on plant immunity, providing new ideas and methods for optimizing soil management and enhancing crop disease resistance [50,51]. In particular, biological control measures, such as the use of beneficial microorganisms to colonize the rhizosphere and induce systemic resistance in plants, have attracted widespread attention due to their advantages, such as environmental friendliness and long-lasting effects [52,53].

Rodríguez-Yzquierdo et al. provided crucial insights into the soil-related factors contributing to the susceptibility of banana crops to Fusarium wilt caused by the soilborne fungus *Fusarium oxysporum* f. sp. *cubense* Tropical Race 4 (Foc TR4) (Syn. *Fusarium odoratissimum*) [54]. To determine the soil-inducing factors of Banana Fusarium wilt TR4, 23 types of soil physical and chemical properties were investigated at 3 commercial banana farms in Laguahira, Colombia. By comparing soil samples from disease-affected plots with those from disease-free areas, researchers identified several key soil attributes associated with Foc TR4 susceptibility. The findings revealed that lower values of pH, cation exchange capacity (ECEC), and saturated hydraulic conductivity (HC), along with reduced levels of organic matter (OM), calcium (Ca), magnesium (Mg), zinc (Zn), and sand, were prevalent in the affected plots. This study highlights the importance of a holistic approach to soil health and disease prevention, offering valuable guidance for banana growers and agricultural researchers alike.

Hao et al. explored the potential of using the attenuated strain *Gibellulopsis nigrescens* Vn-1 to increase potato resistance against Verticillium wilt caused by *Verticillium dahliae* [55]. In this study, the pathogenicity of 72 *V. dahliae* strains was tested to identify the most virulent strain (Vd-36). When potatoes were treated with a spore suspension of *G. nigrescens* Vn-1 at a concentration of 1×10^6 conidia/mL and subsequently infected with *V. dahliae* Vd-36, they exhibited the strongest resistance to Verticillium wilt. The postinoculation responses of potato plants, including the production and accumulation of reactive oxygen species (ROS) and hydrogen peroxide (H_2O_2), as well as changes in related defense enzymes, were examined. The salicylic acid (SA) content in the inoculated plants was measured, and the expression levels of SA biosynthesis-related genes (such as StNPR1 and StPR1b) were analyzed. Twelve hours after inoculation with the attenuated strain Vn-1, significant production and accumulation of ROS and H_2O_2 were observed in potato leaves.

3. Conclusions and Prospects

This special issue comprehensively summarizes the latest advancements in crop fungal pathogenicity research, encompassing biocontrol and antibacterial activity, pathogen interactions and defense mechanisms, genetic diversity and population structure, omics technology, and integrated research methods, as well as soil health and disease control.

The interaction mechanisms between plant pathogens and host plants are complex [56,57]. Pathogens secrete various virulence factors, such as effector proteins, toxins, and metabolites, which interfere with immune signaling pathways in plants, inhibit defense responses,

and eventually lead to disease [58,59]. However, plants have evolved sophisticated immune systems to detect pathogen invasion through pattern recognition receptors, activate intracellular signal transduction, and induce the expression of defense genes to resist infection [60,61]. In the past decade, the development and application of omics technologies have significantly improved our understanding of the molecular mechanisms underlying plant–pathogen interactions [62,63]. Pathogen genomics studies have shown that many important plant pathogenic fungi possess numerous virulence factor genes with diverse functions [64,65]. Comparative genomic analyses have revealed that these virulence genes are usually clustered and differ significantly among species, suggesting their rapid evolution during pathogen adaptation to hosts [66,67]. Further research using advanced omics technologies, such as genomics, transcriptomics, proteomics, and metabolomics, is needed to elucidate the intricacies of these interactions.

Plant growth-promoting rhizobacteria (PGPR) are beneficial bacteria that thrive in the plant rhizosphere and promote plant growth and development through various mechanisms [32,68]. These bacteria form mutually beneficial symbiotic relationships with plant roots and play crucial roles in plant growth, including nitrogen fixation, phosphate solubilization, plant hormone production, the induction of plant resistance, and pathogen inhibition [69,70]. PGPRs show great potential for reducing the use of chemical fertilizers and pesticides, increasing crop yields, and improving soil health [71,72]. The application of modern biotechnologies such as genomics, transcriptomics, and proteomics has provided new perspectives for analyzing the molecular mechanisms of PGPR–plant interactions [73]. Endophytic PGPR can enter and colonize plant tissues, while ectophytic PGPR primarily grow in the soil surrounding plant roots [74,75]. PGPR can synthesize a variety of plant hormones, such as auxin, cytokinin and gibberellin, which directly promote the growth and development of plants [76]. PGPR can also improve plant resistance to pathogens by inducing systemic resistance in plants, thus reducing the occurrence of disease. Future research should further explore the diversity of PGPR and their mechanism of action combined with modern biotechnology to develop more efficient and environmentally friendly agricultural applications to address the challenges faced by global agricultural production [68,72].

Future research of *Agronomy* on crop pathology may include the following areas: (1) to study the genome and molecular mechanism of plant pathogens, reveal the interaction between pathogens and host plants, and provide new targets and strategies for disease control; (2) through traditional breeding and modern biotechnology (such as gene editing, transgenic technology, etc.), cultivate crop varieties with broad spectrum disease resistance and improve the disease resistance of crops; (3) to study the spread and evolution of pathogens in agroecosystems, and explore the application of eco-agriculture and biological control in disease management to reduce the use of chemical pesticides; (4) develop and promote integrated disease management strategies that combine agricultural measures, biological control, chemical control and physical control to achieve sustainable disease control results; (5) study the impact of climate change on crop diseases, including the distribution, spread and pathogenicity of pathogens, to provide a scientific basis for coping with disease risks under climate change.

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